

MASON MENSER

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EDUCATION

Washington State University

August 2022 – May 2026

Bachelor of Science in Computer Science, Minors in Math and History, GPA: 3.81

Pullman, WA

- **Awards:** University Achievement Award, President's Honor Roll (7/7 semesters), John W. and Anna M. Scott Fund for Academic Excellence Scholarship
- **Activities:** Students Book Corporation Treasurer, Sigma Phi Epsilon Scholarship Chair, Men's Club Soccer
- **Coursework:** Embedded Systems, Systems Programming, Databases, Algorithms, Data Structures, Computer Architecture, Software Engineering, Calculus III

EXPERIENCE

Embedded Research Assistant

August 2024 - Present

Washington State University

Pullman, WA

- Sole developer of the control system for an automated temperature-regulated agricultural research enclosure, integrating Raspberry Pi, Arduino, LoRa wireless modules, RTD temperature sensors, and motors
- Implement a PID-style control algorithm in C++ to drive stepper motors regulating a voltage transformer, maintaining real-time temperatures within $\pm 1.0^{\circ}\text{F}$ of target
- Build a remote monitoring and control dashboard using React and Flask on Raspberry Pi 4, containerized with Docker and deployed behind Nginx and Cloudflare Tunnels for secure remote system management

Machine Learning Research Assistant

November 2025 - Present

Washington State University

Pullman, WA

- Develop multilayer perceptron (MLP) models in PyTorch to predict bending stiffness from sandwich panel properties, achieving an R^2 of 0.95
- Implement conditional variational autoencoder (VAE) models to generate geometric properties conditioned on bending stiffness, achieving an R^2 of 0.97 on reconstruction performance
- Co-author preliminary research manuscript with faculty; submission for publication anticipated April 2026

IT Operations Intern

May 2025 - August 2025

PACCAR Parts

Renton, WA

- Implemented a 3rd-party SaaS cloud printing application, enabling native iOS printing and saving over \$50,000 annually in operational cost
- Wrote PowerShell & Python scripts to automate IT department auditing, reducing manual labor by 30 minutes per audit

PROJECTS

OBD2 Vehicle Telemetry Display | C, CAN, SPI, ESP-IDF, FreeRTOS

- Built an embedded telemetry system on ESP32 interfacing with vehicle OBD2 via CAN bus
- Parsed CAN frames to decode standard diagnostic PIDs including RPM and coolant temperature, rendering live diagnostics to an I2C LED display
- Implemented concurrent firmware using FreeRTOS tasks to manage CAN polling and display updates

MashWiki | TypeScript, React Native, Node.js, AWS, OpenAI Embeddings

- Developed a full-stack mobile app recommending Wikipedia articles via vector embeddings and cosine similarity
- Built frontend in React Native and deployed a Node.js backend API on AWS EC2 running PostgreSQL

Peer-to-Peer File Sharing System | C/C++, Linux, Networking, SSL/TLS

- Built a command-line peer-to-peer file sharing and chat application using socket programming in C/C++ on Linux
- Implemented TLS/SSL encrypted communication channels for secure message and file transfer across a local network

TECHNICAL SKILLS

Languages: Python, C++, C, C#, SQL, TypeScript

Frameworks: React, NodeJS, Express, Flask, React Native, Tailwind, Qt

Technologies: ESP-IDF, FreeRTOS, UART, I2C, SPI, Arduino IDE, Raspbian, Git, Linux, PostgreSQL, Docker